



NyayaMithra: A Generative AI-Based Legal Assistance System for Indian Citizens

CSA6502- Generative AI for and Large Language Models

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INTRODUCTION



- Access to legal information is difficult for many Indian citizens due to complex legal language and high consultation costs.
- Most legal resources are available on government portals, making them difficult for common people to understand.
- Citizens often face challenges in understanding legal procedures, documents, and complaint filing.
- NyayaMithra is a Generative AI-based legal assistant that provides simple and reliable legal guidance.
- It uses Retrieval-Augmented Generation (RAG) to retrieve information from official Indian legal documents and generate accurate.





PROBLEM STATEMENT

- Many Indian citizens lack awareness of their legal rights and procedures.
- Legal information is often written in complex language, making it difficult for non-lawyers to understand.
- Hiring legal professionals can be expensive and time-consuming.
- Understanding legal documents and preparing complaints or applications is challenging for common citizens.
- There is a need for an AI-powered legal assistant that provides simple, reliable, and accessible legal guidance using official legal sources.



OBJECTIVES



- Simplify Indian legal information into easy-to-understand language.
- Answer legal questions using official legal documents.
- Analyze and summarize legal documents.
- Generate legal complaint and application drafts.
- Support multilingual interaction.
- Improve legal awareness using Generative AI.





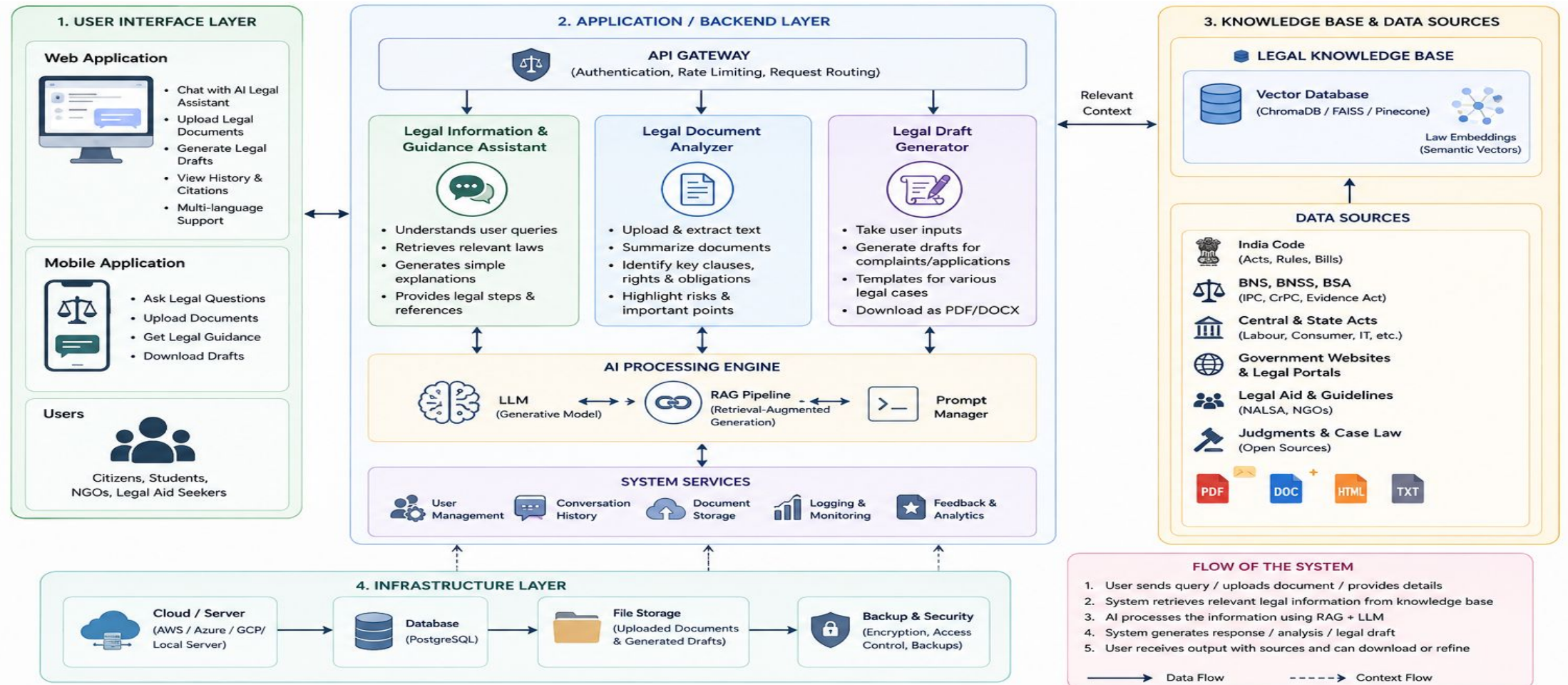
Literature Review



Author Name	Year	Title of Paper	Journal/Conference	Methodology	Limitations
Bommarito II, M. & Katz, D.	2024	GPT-4 Passes the Bar Exam: Artificial Intelligence and Legal Reasoning	Artificial Intelligence & Law	GPT-4, Large Language Models (LLMs)	May generate hallucinated legal information without external knowledge.
Lewis et al.	2022	Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks	NeurIPS	Retrieval-Augmented Generation (RAG), Dense Retrieval	Performance depends on the quality of the retrieved documents.
Xiao Liu et al.	2023	ChatLaw: Open-Source Legal Large Language Model	arXiv	LLM Fine-Tuning, Legal QA, Prompt Engineering	Primarily trained on Chinese legal datasets, limiting direct applicability to Indian law.
Jiang et al.	2024	Large Language Models for Legal Intelligence: A Survey	ACM Computing Surveys	LLMs, RAG, Prompt Engineering	Highlights concerns regarding bias, explainability, and legal reliability.
Zheng et al.	2025	LegalRAG: Retrieval-Augmented Generation for Legal Document Understanding	IEEE Access	RAG, Vector Database, Semantic Search	Requires updated legal knowledge bases for accurate responses.



SYSTEM ARCHITECTURE





Module 1: AI Legal Advisor & Rights Navigator

What it does:

Provides legal guidance by explaining Indian laws, citizen rights, and legal procedures in simple language.

Techniques used:

RAG, Large Language Model (LLM), NLP, Semantic Search, Prompt Engineering.

Components used:

India Code, BNS, BNSS, BSA, Vector Database, Embedding Model, LLM API, FastAPI.

How it works:

- User asks a legal question.
- Relevant legal sections are retrieved.
- AI generates a simple explanation.
- Legal references are displayed.
- User receives guidance with citations.



Module 1: Implementation



NM NyayaMithra
INDIAN LEGAL ASSISTANT

Dashboard

Legal Chat

Document Summarizer

Complaint → Draft

Speech Assistant

Grounded in Indian Law
Statutory guidance adhering to Indian Acts, consumer laws & procedural rules.

Dashboard
AI-Powered Indian Legal Assistant & Drafting System

English

Export Document

Welcome to NyayaMithra

Understand your legal problem in simple language.

What would you like to do?

Ask a Legal Question

Summarize a Document

Create a Complaint / Legal Draft

Use Voice

Ask

Get simple explanations of Indian laws.

Understand

Upload a legal document and understand it easily.

Draft

Turn your complaint into a preliminary legal draft.

Disclaimer: This is an AI-assisted legal draft for informational purposes only. Consult a legal professional for formal representation.

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Module 2: Smart Legal Document Intelligence



What it does:

Analyzes legal documents, summarizes content, and explains important clauses in simple language.

Techniques used:

Document Parsing, Text Summarization, RAG, NLP, OCR (Optional).

Components used:

PDF Upload, PyMuPDF, Embedding Model, Vector Database, LLM, FastAPI.

How it works:

- User uploads a legal document.
- Text is extracted and processed.
- AI summarizes the document.
- Important clauses and risks are identified.
- Simplified results are displayed.



Module 2: Implementation



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Document Summarizer
Upload a legal document and NyayaMithra will explain it in simple language

English

Export Document

Document Summarizer
Upload a legal document and NyayaMithra will explain it in simple language.

English

Click to upload or drag and drop a legal document
Supported formats: PDF, DOCX, TXT
nyayamithra_sample_legal_notice.pdf

Or paste the document text below: Clear

[Document File Loaded: nyayamithra_sample_legal_notice.pdf (23 KB)]

THIS RENTAL AGREEMENT is made on this 1st day of April 2026 between Sri Rajesh Sharma (Landlord) and Sri Priya Patel (Tenant).
The Landlord hereby lets and the Tenant hereby takes the residential flat located at Flat 402, Green Valley Apartments, Bengaluru for a monthly rent of Rs. 28,000/-.
The Tenant has deposited an interest-free security deposit of Rs. 1,50,000/-

Summarize Document

Disclaimer: This is an AI-assisted legal draft for informational purposes only. Consult a legal professional for formal representation.

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Module 3: AI Legal Action & Draft Assistant

What it does:

Generates legal complaint and application drafts based on user-provided information.

Techniques used:

Prompt Engineering, LLM, Natural Language Generation (NLG), Template-Based Generation.

Components used:

Draft Templates, User Forms, LLM API, Document Generator (PDF/DOCX), FastAPI.

How it works:

- User selects a document type.
- Required details are entered.
- AI generates the legal draft.
- User reviews the draft.
- Document is downloaded or edited.



Module 3: Implementation

NM NyayaMithra
INDIAN LEGAL ASSISTANT

Dashboard

Legal Chat

Document Summarizer

Complaint → Draft

Speech Assistant

Grounded in Indian Law

Statutory guidance adhering to Indian Acts, consumer laws & procedural rules.

Complaint → Legal Draft Generator

Describe your dispute or grievance in plain words. NyayaMithra extracts the legal facts and drafts a court-ready notice or complaint.

English

Export Document

1. DESCRIBE WHAT HAPPENED IN SIMPLE WORDS

Voice Input

My landlord has not returned my security deposit of ₹1,50,000 even though I left the house two months ago on 15 March 2026. The house was handed over in perfect condition and all bills were paid.

Select Draft Type:

Legal Notice

Consumer Complaint

Police Complaint

RTI Application

General Application

Generate Preliminary Draft

2. KEY DETAILS (EDIT ANYTIME TO REFINE DRAFT)

Your Name

Priya Patel

Amount involved (₹)

₹1,50,000

Your Address / City

Flat 101, Palm Grove, Indiranagar, Bengaluru - 560038

Opposite Party (Company / Landlord / Person)

Rajesh Sharma (Landlord), #42 Green Valley Apartments, Bengaluru

Preliminary Legal Draft

Format: Legal Notice

Read Aloud

Copy

Download

Print

LEGAL NOTICE

To,
Rajesh Sharma,
#42 Green Valley Apartments,
Bengaluru, Karnataka.

Dated: May 20, 2026

SUBJECT: Formal Demand Notice for Immediate Refund of
Security Deposit of ₹1,50,000/- with Interest and Damages.

Sir / Madam,

Under instructions from and on behalf of my client / the undersigned, Priya Patel, residing at Flat 101, Palm Grove, Indiranagar, Bengaluru - 560038, I hereby serve upon you this formal Legal Notice:

Note for Citizens:
This is a preliminary draft to help you organize facts and legal grounds. Always review details carefully.

Disclaimer: This is an AI-assisted legal draft for informational purposes only. Consult a legal professional for formal representation.

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Future Enhancements

- **Multilingual Support** – Extend assistance to major Indian regional languages.
- **Voice-Based Legal Assistance** – Enable users to ask legal queries using voice.
- **Real-Time Legal Updates** – Integrate updated Indian laws, acts, and court judgments.
- **Personalized Legal Guidance** – Provide responses based on the user's specific case details.
- **Legal Document Automation** – Automatically generate petitions, applications, notices, and complaints.
- **Court & Case Tracking** – Provide case-status and hearing-date information through integration with official sources.
- **Advanced RAG System** – Improve accuracy by retrieving information from verified legal databases.
- **Mobile Application** – Develop an Android/iOS app for easier access to legal assistance.



Challenges Faced

- **Legal Data Availability** – Collecting reliable and updated Indian legal documents.
- **Accuracy of Responses** – Generative AI may sometimes produce incorrect or misleading information.
- **Complex Legal Language** – Converting complicated legal terminology into simple language.
- **Multilingual Complexity** – Maintaining accuracy across different Indian languages.
- **Data Privacy** – Protecting sensitive personal and legal information provided by users.
- **Hallucination Risk** – AI may generate information that is not supported by legal sources.
- **Frequent Law Changes** – Keeping the knowledge base updated with amendments and new judgments.
- **Legal Responsibility** – Ensuring the system provides assistance without replacing professional legal advice.



CONCLUSION



- NyayaMithra leverages Generative AI and Retrieval-Augmented Generation (RAG) to make legal information simple, accessible, and reliable for Indian citizens.
- The system provides legal guidance, document analysis, and legal draft generation using official legal sources.
- It promotes legal awareness, reduces the complexity of legal procedures, and improves access to legal information.
- The platform is designed as an informational and educational assistant, supporting—not replacing—professional legal services.
- Future enhancements include voice support, regional language expansion, real-time legal updates, and integration with government legal portals.

